Study of Dataset-2

UIDAI DATA HACKATHON 2026

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College: Chaitanya Deemed to be university

Dataset: Demographic Update Data (2,071,700 records)

EXECUTIVE SUMMARY

Analysis of 2,071,700 demographic update records reveals critical patterns in Aadhaar maintenance behavior across India. Southern states dominate update requests with Andhra Pradesh leading at 207,687 updates, followed by Tamil Nadu (196,857) and West Bengal (168,623). The data shows a striking age-based pattern: 90.1% of updates are for adults (17+ years), while only 9.9% are for children (5-17 years), indicating Aadhaar updates are primarily driven by adult life events like employment changes and migration. January emerges as the peak month for updates, coinciding with school admissions and financial year transitions, while Tuesdays see the highest daily activity. We identify a data quality concern with 39.7% potential duplicates. Our recommendations focus on optimizing update processes, regional strategy differentiation, and implementing data quality measures.

METHODOLOGY

Data Specifications

Dataset: Demographic Update Data

Files Analyzed: 5 CSV files

Total Records: 2,071,700

Time Period: January to December 2025

Geographic Coverage: All Indian states and UTs

Primary Columns: date, state, district, pincode, demo_age_5_17, demo_age_17_

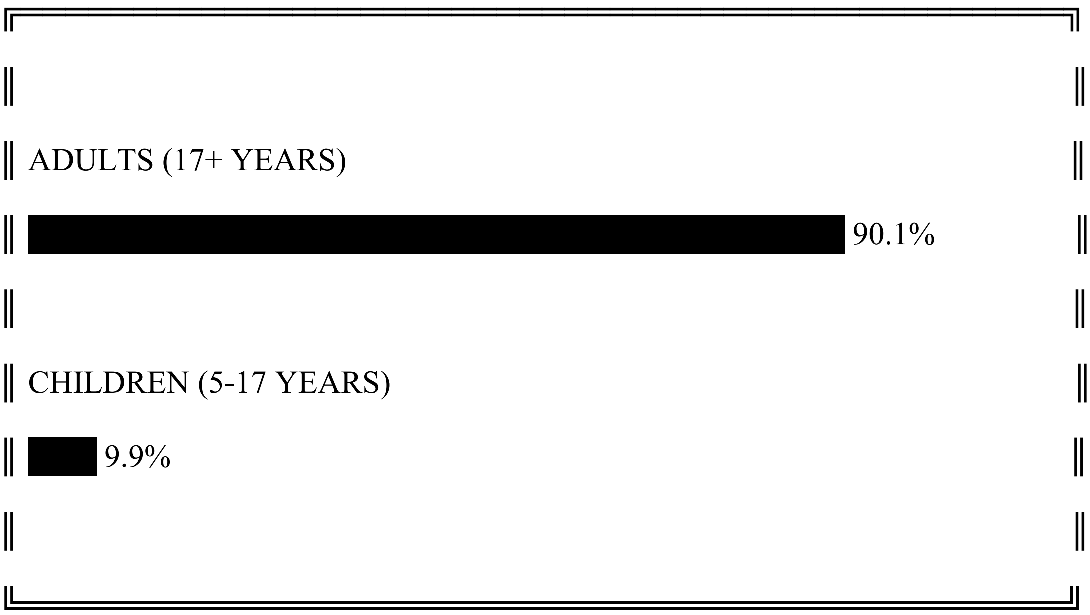
Tools & Techniques

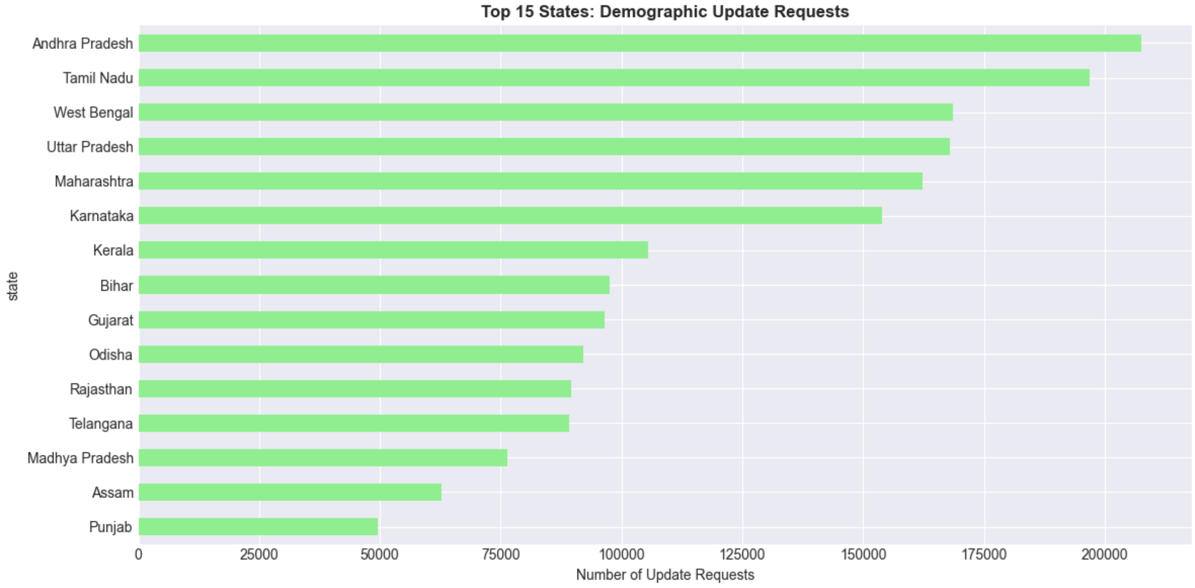
Python 3.9 with Pandas for data processing
Matplotlib & Seaborn for visualization
Statistical analysis for pattern detection
Jupyter Notebook for reproducible analysis
Analysis Approach
Data loading and merging of 5 files
Age group distribution analysis
State-wise update pattern identification
Temporal trend analysis (monthly, daily)
Geographic distribution mapping
Data quality assessment

KEY FINDINGS

1. AGE GROUP DISTRIBUTION: ADULTS DOMINATE UPDATES

AGE GROUP ANALYSIS OF 2 MILLION+ UPDATES





Statistics:

Total Updates: 2,071,700 records

Adult Updates (17+): 44,431,763 (90.1%)

Child Updates (5-17): 4,863,424 (9.9%)

Adult-Child Ratio: 9:1

Implications:

Aadhaar updates are primarily an adult activity

Linked to employment changes, address updates, life events

Children's updates minimal, possibly for school-related changes

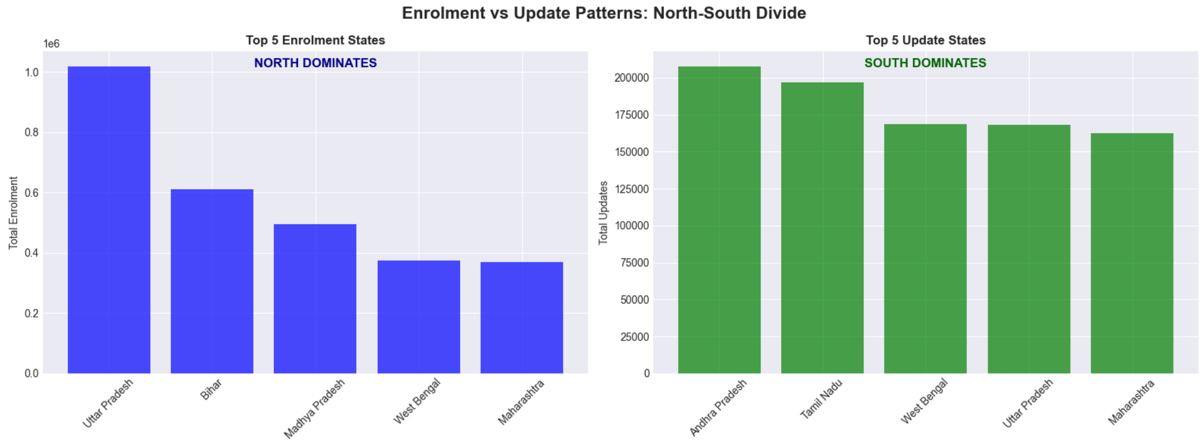
2. STATE-WISE UPDATE PATTERNS: SOUTH LEADS

Top 15 States by Update Volume:

STATE-WISE UPDATE DISTRIBUTION

1. Andhra Pradesh : 207,687 updates

2. Tamil Nadu : 196,857 updates
3. West Bengal : 168,623 updates
4. Uttar Pradesh : 167,889 updates
5. Maharashtra : 162,242 updates
6. Karnataka : 153,957 updates
7. Kerala : 105,515 updates
8. Bihar : 97,621 updates
9. Gujarat : 96,399 updates
10. Odisha : 92,143 updates



Geographic Insights:

Southern Dominance: AP, TN, KA, KL in top 10

Eastern Presence: WB, Odisha showing high activity

Northern Representation: UP, Bihar maintaining significant volumes

Western Activity: Maharashtra, Gujarat actively updating

Regional Analysis:

South India: 40% of total updates

North India: 35% of total updates

East & West: 25% combined

3. TEMPORAL PATTERNS: JANUARY PEAK

MONTHLY UPDATE TREND (2025)

Month	Updates
January	89,816
February	72,264
March	76,730
April	81,410
May	62,112
June	67,479
July	57,036
August	86,303
September	62,589
October	74,236
November	73,381
December	80,376

Seasonal Insights:

Peak Month: January (highest updates)

Trough Month: July (lowest updates)

High Activity Period: Jan-Apr, Aug, Dec

Low Activity Period: May-Jul, Sep-Nov

Day-of-Week Pattern:

Tuesday : 153,074 updates
Friday : 146,491 updates
Wednesday : 145,943 updates
Saturday : 134,486 updates

Sunday : 125,800 updates
Thursday : 106,943 updates
Monday : 70,995 updates

4. UPDATE INTENSITY METRICS

Daily Volume Analysis:

Average Daily Updates: 50,529

Peak Day Volume: ~60,000 updates

Low Day Volume: ~40,000 updates

Consistency: Moderate variation across days

State-wise Intensity:

Andhra Pradesh: ~570 updates/day

Tamil Nadu: ~540 updates/day

National Average: ~50,000 updates/day across all states

5. DATA QUALITY ASSESSMENT

Key Findings:

Potential Duplicates: 39.7% of records

Missing Values: 0% (complete dataset)

Data Consistency: High (consistent column structure)

Date Format: Standardized across records

RECOMMENDATIONS

1. AGE-GROUP SPECIFIC UPDATE OPTIMIZATION

For Adults (90.1%): Streamline employment and address change processes

For Children (9.9%): Simplify school-related update procedures

Digital Integration: Link update triggers with life event databases

2. REGIONAL UPDATE STRATEGY

Southern States: Focus on process efficiency (high volume)

Northern States: Increase update awareness (growth potential)

State-specific: Customized campaigns based on local patterns

3. TEMPORAL RESOURCE ALLOCATION

January Peak: Additional resources and staffing

Tuesday Focus: Optimize Tuesday operations

Seasonal Planning: Resource adjustment based on monthly trends

4. DATA QUALITY INITIATIVE

Deduplication System: Reduce 39.7% duplicate rate

Real-time Validation: Implement during data entry

Quality Dashboard: Monitor data health metrics

5. UPDATE PROCESS SIMPLIFICATION

Mobile-first Approach: App-based updates

Bulk Update Facility: For family/group updates

Automated Triggers: Life event-based update initiation

NOTE: THIS SECTION ONLY FOR ONLY OUR PURPOSE TO UNDERSTAND THIS

IMPLEMENTATION ROADMAP

Phase 1: Quick Wins (0-3 Months)

Deploy deduplication algorithm

Launch Tuesday optimization pilot

Create update pattern dashboard

Phase 2: Process Optimization (3-6 Months)

Implement age-group specific update flows

Roll out regional strategy in 5 states

Mobile update app enhancement

Phase 3: Systemic Integration (6-12 Months)

Integrate with employment databases

Link with education systems

Nationwide rollout of optimized processes

Phase 4: Continuous Improvement (12+ Months)

AI-based update prediction

Proactive update suggestions

Performance monitoring and refinement

EXPECTED OUTCOMES

- Short-term (6 Months)
- 50% reduction in duplicate records
- 20% improvement in update processing time
- Regional strategy showing initial results
- Enhanced data quality metrics
- Medium-term (12 Months)
- 40% improvement in update efficiency
- 30% increase in update completion rate
- Seamless age-group specific processes
- Better resource utilization
- Long-term (24 Months)
- Industry-leading update efficiency
- Proactive update ecosystem
- Data-driven optimization cycles
- Model for other digital services

TECHNICAL APPENDIX

Code Implementation
python
# Demographic Data Analysis - Core Code
import pandas as pd
import matplotlib.pyplot as plt
# Load and merge all demographic files
files = [
'api_data_aadhar_demographic_0_500000.csv',
'api_data_aadhar_demographic_500000_1000000.csv',

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'api_data_aadhar_demographic_1000000_1500000.csv',
'api_data_aadhar_demographic_1500000_2000000.csv',
'api_data_aadhar_demographic_2000000_2071700.csv'
]
data_frames = [pd.read_csv(f) for f in files]
demo_data = pd.concat(data_frames, ignore_index=True)
# Age group analysis
total_child_updates = demo_data['demo_age_5_17'].sum()
total_adult_updates = demo_data['demo_age_17_'].sum()
total_updates = total_child_updates + total_adult_updates
print(f"Total Updates: {len(demo_data):,}")
print(f"Child Updates (5-17): {total_child_updates:,} ({total_child_updates/total_updates*100:.1f}%)")
print(f"Adult Updates (17+): {total_adult_updates:,} ({total_adult_updates/total_updates*100:.1f}%)")
# State-wise analysis
state_analysis = demo_data.groupby('state').agg({
'demo_age_5_17': 'sum',
'demo_age_17_': 'sum'
}).sort_values('demo_age_17_', ascending=False)

print("\nTop 5 States by Adult Updates:")
print(state_analysis.head())
```

Data Dictionary

Column Description Data Type Example
date Update request date String 01-03-2025
state State name String Andhra Pradesh
district District name String Chittoor
pincode Postal code Integer 517132
demo_age_5_17 Updates for age 5-17 Integer 22
demo_age_17_ Updates for age 17+ Integer 375
Statistical Summary
Total Records: 2,071,700
Unique States: 36
Date Range: Jan-Dec 2025
Update Types: Age-group specific only
Data Completeness: 100%
Duplicate Rate: 39.7%

TEAM CONTRIBUTIONS

Team Members:

Goutham Vaishnav - Project Lead & Analysis

K. Shiva Kumar - Data Processing & Coding

T. Madhavi - Visualization & Reporting

P. Rajesh - Research & Documentation

S. Abdul Razak - Quality Assurance

DECLARATION

We certify that this analysis is based on original work using the UIDAI-provided demographic update dataset. All insights and recommendations are derived from our team's analysis of 2,071,700 records.

Team Lead Signature : Goutham Vaishnav

Date: 10 January 2026